Sahaj Bhadja

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Summary

Embedded Software Engineer with 2+ years applying **C/C++ and Python**, **simulation-driven testing**, CI/CD, debugging, and control-system design.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Object-Oriented Programming, Software Engineering, Computer Networks

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built fault-tolerant Python services with explicit job states, timeouts, failure isolation, and monitoring, processing 1K+ jobs at 3x throughput while strengthening **deterministic behavior and production reliability across services**.
- Architected multi-tenant Azure SQL and storage workflows for 500+ users, reducing latency 71% through **structured debugging, indexed queries, data validation, and repeatable performance testing across production workloads**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Engineered concurrent Python processing with ThreadPoolExecutor, OCR, timeouts, and model fallbacks, handling 1K+ files per batch while reducing turnaround 60% through **functional testing and fault isolation during peak load**.
- Validated React and Flask releases supporting 10K+ daily requests, reproducing production defects and coordinating deployments through **unit tests, integration testing, code review, and documented troubleshooting workflows across environments**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Built a PyTorch Decision Transformer for autonomous-driving simulation, processing 100K+ trajectory logs through reproducible pipelines and improving planning 18% across **vehicle-operation modeling and controlled simulation testing for autonomy experiments**.
- Integrated HD-map geometry, lane direction, and ego trajectories with spatial-temporal encoding, improving closed-loop stability 15% while **debugging state transitions, control behavior, and simulation edge cases in research baselines**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Designed object-oriented modules using Singleton and Factory patterns, improving API response times 30% while **troubleshooting integrations across Windows, Linux, macOS, AWS services, and cross-platform execution environments during release cycles**.
- Automated builds and 20+ unit and integration suites with Jenkins, PyTest, Mockito, and virtual machines, increasing pipeline efficiency 68% through **continuous integration and reproducible release validation across platforms**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **well-defined interfaces, state-aware workflows, and reliable multi-system communication across distributed production environments**.
- Automated Docker packaging and Jenkins releases, cutting deployment time 50% while using Sentry, Azure Monitor, and Kibana for **instrumentation, regression investigation, and dependable software releases across environments**.

Skills

Core: C, C++, Python, Java, Bash, MATLAB, Simulink, Stateflow, Data Structures, Algorithms, OOP
Embedded: Firmware, CAN, J1939, SocketCAN, State Machines, Real-Time Systems, Control Systems, Controller Interfaces, Hardware Schematics, Requirements Traceability
Testing: Unit Testing, Integration Testing, Functional Testing, Bench Testing, SIL, HIL, Vehicle Simulation, Fault Injection, Regression Testing, GoogleTest, PyTest
Tools: Git, Linux, CMake, GDB, Jenkins, CI/CD, Docker, Agile, Debugging, Design Documentation, Technical Communication

Projects

Vehicle Controller HIL Bench | *C++, CAN J1939, SocketCAN, GoogleTest, Python*
- Developed **embedded vehicle-control software** with C++ state machines, CAN J1939 messaging, sensor plausibility checks, diagnostics, and fail-safe transitions, validated through unit tests, controller bench scenarios, and HIL fault injection.

Machine Control Model | *MATLAB, Simulink, Stateflow, SIL, HIL, CI/CD*
- Designed a **model-based machine-control application** from requirements and state diagrams, generating testable control logic and validating operating modes, interlocks, and fault responses through SIL/HIL regression suites before release.

CPU-Scheduling Simulator | *C++, React, JavaScript, Algorithms* | Live
- Implemented **FCFS, SJF, SRT, Round Robin, priority, and multilevel scheduling** in C++, validating correctness and visualizing execution behavior across 1K+ simulated process combinations with interactive controls during testing.